McMaster University

Mathematics and Statistics

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We recognize and acknowledge that McMaster University meets and learns on the traditional territories of the Mississauga and Haudenosaunee nations, and within the lands protected by the "Dish With One Spoon" wampum, an agreement amongst all allied Nations to peaceably share and care for the resources around the Great Lakes.

STATS 3Y03 –PROBABILITY AND STATISTICS FOR ENGINEERING

2025 WINTER TERM

COURSE ADMINISTRATION:

Course Instructor: **Dr. Angelo Canty. Hamilton Hall 209. cantya@mcmaster.ca**

Class Times: Monday, 12:30pm–1:20pm; Tuesday, 1:30pm–2:20pm; Thursday, 12:30pm–1:20pm.
Class Locations: See information on mosaic or Avenue.
Office Hours: Monday, 1:30pm–2:30pm; Tuesday 12:00pm–1:00pm; Thursday, 1:30pm–2:30pm.
Course Website: The course home page will be on Avenue to Learn. Class notes, assignments and other important information will be posted there so check regularly.

COURSE DESCRIPTION

Introduction to probability, data analysis, statistical inference, regression, correlation and analysis of variance.
Prerequisite: Registration in Level II or above of a program in Engineering above Level I Computer Science, Computer Engineering, Electrical Engineering, Mechanical Engineering, Mechatronics or Software Engineering
Antirequisite: STATS 3J04

MATERIALS & FEES

Required Materials/Resources

Textbook

1. *Applied Statistics and Probability for Engineers, 7th Edition* by Montgomery and Runger, Wiley. (Online edition available through the bookstore, or wherever you can find it).

Optional Materials/Resources

Textbooks

1. Student Solutions Manual for the required textbook
2. "40 Puzzles and Problems in Probability and Mathematical Statistics" - Wolfgang Schwarz (available for free download on McMaster SpringerLink)

Software

1. R available for free download at https://cran.r-project.org/
2. RStudio available for free download at https://posit.co/downloads/

COURSE OVERVIEW AND ASSESSMENT

A selection of topics will be covered from the course textbook. A tentative list is as follows:

Topic
Introduction to probability
Conditional probability and independence
Discrete random variables
Continuous random variables
Bivariate random vectors
Descriptive statistics
Point estimation
Confidence intervals for means and proportions
Hypothesis tests for means and proportions
Inference for two samples
Simple linear regression
Analysis of variance

Software: Throughout the course I will use the R software for analysis of data. I will post any code used (appropriately commented) on Avenue after the class so that students may try the examples on their own. Students will not be required to learn how to use R, however, and it will not be tested on any tests although students may be required to make conclusions from R output.

Assignments: There will be five assignments each of which will have 4 short-answer questions. Additional information about assignment format, submission (Crowdmark) and due dates will be announced on the course website well in advance. The best 4 of 5 assignments will count towards your final grade.

Term Tests: There will be two term tests. Each test will have some multiple choice questions and short answer questions. **The tests will take place in the evening at 7:00-8:15pm.** The dates below are tentative and subject to change. Term test days/times will be confirmed in class and on the course website well in advance. The location and more information about the tests will also be announced in class and posted on the course website at least 2 weeks in advance of the test date.

Tentative Term Test Dates:

Test	Test Date
Term Test #1	Thursday, February 13, 2025
Term Test #2	Thursday, March 27, 2025

Final Exam: There will be a cumulative final exam during the April exam period. The format of the exam will be similar to the tests (but twice as long). More information will be posted on the website and announced in class before the end of the semester.

Calculator: McMaster standard calculator is allowed for the tests and the final exam.

Formula Sheet: A formula sheet will be provided as part of the tests and exam. This sheet will be posted on the course website in advance of the test/exam.

McMaster Photo ID: For the tests and the final exam, students are **required** to bring their McMaster photo ID cards for verification purposes.

COURSE EVALUATION

The final grade will be based on the following components:

Grade Component	Scheme 1 weight	Scheme 2 weight
Best 4 of 5 Assignments	20% (5% each)	20% (5% each)
2 Tests	40% (20% each)	30% (best test)
Final Exam	40%	50%

Both weightings will be calculated for all students who sit both term tests and the final grade will be based on the better of the two marks. At the end of the course, the weightings used and final grades may be adjusted, but this can only increase the final grade and will be done uniformly. The grade equivalence chart published in the Undergraduate Calendar will be used to convert between percentages and letter grades.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

- Any student who misses work during the term should complete the online McMaster Student Absence Form (MSAF) within one week of the missed work. Second or subsequent absences must be approved by the associate dean of your faculty. For more details see www.mcmaster.ca/msaf
- In the event of a student using an MSAF for absence at one of the two term tests, only the second marking scheme listed above will be used automatically. Missing a test can never increase your mark since you will get the best of the two marking schemes in any case.
- **Students may not miss both tests except in extreme documented circumstances and will be required to discuss their options with the course instructor in such a situation.**
- In the case of one or more missed assignments, the assignment mark will be calculated using **all** other assignments not covered by an MSAF or otherwise approved. The assignments used will be equally weighted in calculation of the assignment mark.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's *Academic Accommodation of Students with Disabilities* policy.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the *Academic Integrity Policy*, located at https://secretariat.mcmaster.ca/university-policies-procedures- guidelines/

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to the McMaster Office of Academic Integrity's webpage.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all our living, learning and working communities. These expectations are described in the *Code of Student Rights & Responsibilities* (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.

Please Note: Course Changes

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. **It is the responsibility of the student to check their McMaster email and the course website on Avenue to Learn regularly during the term and to note any changes.** Announcements will be made in class and on Avenue to Learn. Important changes will also be emailed to the class distribution list which uses the student's mcmaster.ca email.